

Spectral Fingerprints of Synthesis: Detecting AI-Generated Images via Two-Dimensional Fourier Analysis

Charitha Ranasinghe
Frequency-Domain Analysis of 2D Signals

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Abstract

The proliferation of photorealistic images synthesised by deep generative models — including Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and diffusion-based architectures — poses a significant challenge to digital media forensics. This paper presents a principled, frequency-domain framework for discriminating AI-generated imagery from natural photographs. We demonstrate that the two-dimensional Discrete Fourier Transform (2-D DFT) exposes characteristic spectral anomalies in synthetic images: elevated high-frequency energy, depressed DC bias, suppressed $1/f^2$ roll-off, and periodic grid-like peaks attributable to convolutional strides and attention tiling artefacts. From the centred log-magnitude spectrum we derive a compact 515-dimensional feature vector through polar frequency binning, spectral flatness (Wiener entropy), and a high-to-low-frequency energy ratio. An ensemble classifier trained on open-source image libraries achieves competitive discriminative performance, providing an interpretable, computationally lightweight alternative to black-box deep-learning detectors.

Keywords: image forensics, Fourier analysis, AI-generated image detection, spectral fingerprinting, generative models, frequency-domain features.

1. Introduction

The advent of large-scale generative models has fundamentally altered the landscape of digital imagery. Systems such as Stable Diffusion [1], DALL-E [2], and StyleGAN [3] can produce photorealistic images indistinguishable to the human eye from genuine photographs. This capability presents a dual threat: erosion of epistemic trust in visual media and facilitation of disinformation at unprecedented scale.

Existing detection methods broadly fall into two categories. *Spatial-domain* approaches analyse pixel-level statistics, noise residuals, or texture inconsistencies [4, 5]. *Deep-feature* approaches fine-tune large convolutional neural networks as binary classifiers [6], achieving high accuracy but sacrificing interpretability and generalisation across generator architectures.

We advocate a third paradigm: *frequency-domain forensics*. The two-dimensional Fourier transform decomposes an image into its constituent spatial frequencies. Natural photographs obey well-characterised statistical laws in the frequency domain — most notably the $1/f^\alpha$ spectral law [9, 10] — which generative models systematically violate. These violations constitute *spectral fingerprints*: stable, generator-agnostic signatures that persist even after JPEG compression, resizing, or mild post-processing.

The contributions of this work are:

- (i) A rigorous derivation of the spectral anomalies introduced by convolutional, attention-

based, and upsampling components of contemporary generative architectures;

- (ii) A novel polar frequency binning scheme that converts the 2-D spectrum into an interpretable, fixed-length feature vector;
- (iii) An open-source detection pipeline connecting to large image libraries (CIFAR-10, DiffusionDB) for reproducible evaluation.

2. Background and Related Work

2.1 Statistical Properties of Natural Images

Natural images are not random; they obey a remarkably consistent statistical regularity in the frequency domain. Let $\mathcal{P}(\rho)$ denote the radially averaged power spectral density at spatial frequency ρ . Field [9] showed empirically and Burton and Moorhead [10] confirmed theoretically that for an ensemble of natural scenes:

$$\mathcal{P}(\rho) \propto \rho^{-\alpha}, \quad \alpha \approx 2, \quad (1)$$

a relationship known as the $1/f^2$ *spectral law* or *pink-noise* characteristic. This implies that low spatial frequencies carry the vast majority of image energy, and power decays monotonically as a power law.

A corollary is that natural images exhibit strong DC bias: the zero-frequency component $F[0, 0] = HW\bar{I}$ — proportional to mean intensity — accounts for a substantial fraction of total spectral energy. Any systematic deviation from (1) is therefore diagnostic of non-natural image generation.

2.2 Spectral Artefacts of Generative Models

Frank *et al.* [5] were among the first to systematically document spectral anomalies in GAN-generated images, observing pronounced peaks in the high-frequency region of the magnitude spectrum. Dzanic *et al.* [7] demonstrated that these artefacts arise from the upsampling operations (bilinear, transposed convolution) intrinsic to generator architectures: upsampling by integer factor s introduces periodic aliases at

multiples of the Nyquist frequency $1/s$, manifesting as a regular grid of bright spots in the centred magnitude spectrum.

Diffusion models [1] exhibit a related but distinct pattern: the iterative denoising process, implemented via a U-Net with attention layers of fixed tile size T , introduces periodicity at frequency $1/T$, producing checkerboard-like artefacts of a different spatial scale than GAN upsampling ghosts.

2.3 Limitations of Deep-Feature Detectors

CNN-based detectors [6, 8] achieve near-perfect in-distribution accuracy but generalise poorly: a detector trained on ProGAN outputs may fail on StyleGAN3 or Stable Diffusion images, since the spatial-domain features exploited differ across architectures. Frequency-domain features are more *architecture-agnostic*: any generative model that incorporates strided convolution, transposed convolution, or attention with fixed context windows will leave spectral traces, regardless of the specific generator design.

3. Theoretical Framework

3.1 The Two-Dimensional Discrete Fourier Transform

Definition 3.1 (2-D DFT). Let $\mathbf{I} \in \mathbb{R}^{H \times W}$ be a discrete grayscale image with pixel values $I[m, n]$. The *two-dimensional Discrete Fourier Transform* is the mapping $\mathcal{F}_{2D} : \mathbb{R}^{H \times W} \rightarrow \mathbb{C}^{H \times W}$ defined by

$$F[u, v] = \sum_{m=0}^{H-1} \sum_{n=0}^{W-1} I[m, n] e^{-j2\pi(\frac{mu}{H} + \frac{nv}{W})}, \quad (2)$$

for $(u, v) \in \{0, \dots, H-1\} \times \{0, \dots, W-1\}$.

The index pair (u, v) represents cycles per image in the vertical and horizontal directions. The DC component at $(u, v) = (0, 0)$ equals the sum of all pixel values:

$$F[0, 0] = \sum_{m, n} I[m, n] = HW\bar{I}. \quad (3)$$

3.2 Centring via Frequency Shift

The standard DFT places the DC component at the corner of the output array. Applying the shift theorem — multiplication of the spatial image by $(-1)^{m+n} = e^{j\pi(m+n)}$ — translates the spectrum by half its extent:

$$I[m, n] (-1)^{m+n} \xrightarrow{\mathcal{F}_{2D}} \hat{F}[u, v] = F\left[u - \frac{H}{2}, v - \frac{W}{2}\right]. \quad (4)$$

The centred spectrum $\hat{F}$ places the DC component at $(\lfloor H/2 \rfloor, \lfloor W/2 \rfloor)$, enabling radial analysis with respect to a geometrically meaningful origin.

3.3 Log-Scaled Magnitude Spectrum

Definition 3.2 (Log Magnitude Spectrum). The *log-scaled magnitude spectrum* is defined as

$$\mathcal{L}[u, v] = 20 \log\left(1 + \left|\hat{F}[u, v]\right|\right), \quad (5)$$

where the additive constant 1 ensures numerical stability at zero-magnitude coefficients, and the factor 20 yields a decibel-like unit.

The transformation (5) is necessary because the dynamic range of $|\hat{F}|$ spans roughly six orders of magnitude between the DC peak and high-frequency noise floor; without compression, the DC component visually dominates and all other spectral structure is invisible.

3.4 The Power Spectrum and Parseval's Identity

The *power spectral density* is $\mathcal{P}[u, v] = \left|\hat{F}[u, v]\right|^2$. By Parseval's theorem, total spatial-domain energy equals total frequency-domain energy up to normalisation:

$$\sum_{m, n} |I[m, n]|^2 = \frac{1}{HW} \sum_{u, v} |F[u, v]|^2. \quad (6)$$

3.5 Spectral Anomalies in Synthetic Images

We formalise the three key spectral deviations introduced by generative architectures.

3.5.1 Violation of the $1/f^2$ Law

Define polar coordinates centred at the DC position:

$$\rho[m, n] = \sqrt{\left(n - \frac{W}{2}\right)^2 + \left(m - \frac{H}{2}\right)^2}. \quad (7)$$

The radially averaged power at radius ρ is:

$$\bar{\mathcal{P}}(\rho) = \frac{1}{|\mathcal{A}_\rho|} \sum_{\{(m, n): \lfloor \rho[m, n] \rfloor = \rho\}} \mathcal{P}[m, n]. \quad (8)$$

Natural images satisfy $\bar{\mathcal{P}}(\rho) \propto \rho^{-2}$. AI images exhibit a *flatter* radial profile — excess energy in mid-to-high frequency bands — quantifiable via the slope deviation:

$$\Delta\alpha = \hat{\alpha}_{\text{natural}} - \hat{\alpha}_{\text{AI}}, \quad \hat{\alpha} = -\frac{\partial \log \bar{\mathcal{P}}}{\partial \log \rho}. \quad (9)$$

3.5.2 Periodic Grid Artefacts

A transposed convolution with stride s performs upsampling equivalent to inserting $(s - 1)$ zeros between activations, then applying a learned filter. Zero-insertion in the spatial domain corresponds to spectral replication at intervals of $1/s$ in the frequency domain (the aliasing theorem):

$$\text{stride-}s \text{ upsampling} \implies \text{peaks in } \mathcal{L} \text{ at } \left\{ \frac{k}{s} \cdot F_{\max} \right\}_{k=1}^{s-1}. \quad (10)$$

Attention layers with tile size T introduce analogous periodicity at $1/T$. These spectral peaks form a visible grid pattern in the centred magnitude spectrum that is absent from natural images.

3.5.3 Suppressed DC Bias

The DC energy fraction,

$$\eta_{\text{DC}} = \frac{\mathcal{P}\left[\frac{H}{2}, \frac{W}{2}\right]}{\sum_{u, v} \mathcal{P}[u, v]}, \quad (11)$$

is consistently lower in AI-generated images. This reflects the loss of global illumination coherence: generative models optimise local patch statistics rather than global scene-level mean intensity, effectively distributing energy more uniformly across the spectrum.

4. Feature Engineering

4.1 Polar Frequency Binning

We partition the centred spectrum into a polar grid with N_r radial rings and N_s angular sectors. Let

$$\Delta\rho = \frac{R_{\max}}{N_r}, \quad R_{\max} = \min\left(\frac{H}{2}, \frac{W}{2}\right), \quad (12)$$

$$\Delta\varphi = \frac{2\pi}{N_s}, \quad (13)$$

and define the angular coordinate $\varphi[m, n] = \arctan\left(\frac{m-H/2}{n-W/2}\right) + \pi \in [0, 2\pi)$.

Each bin $\mathcal{B}_{ij}$ collects spectrum pixels in ring i , sector j :

$$\mathcal{B}_{ij} = \{(m, n) \mid i\Delta\rho \leq \rho[m, n] < (i+1)\Delta\rho, j\Delta\varphi \leq \varphi[m, n] < (j+1)\Delta\varphi\}. \quad (14)$$

The feature for bin (i, j) is the mean log-power within that bin:

$$\phi_{ij} = \frac{1}{|\mathcal{B}_{ij}|} \sum_{(m,n) \in \mathcal{B}_{ij}} \mathcal{L}[m, n]. \quad (15)$$

With $N_r = 32$ and $N_s = 16$ this yields 512 polar features.

4.2 Spectral Flatness (Wiener Entropy)

The *spectral flatness measure* [14] quantifies the degree to which the power spectrum resembles white noise:

$$\gamma = \frac{\exp\left(\frac{1}{HW} \sum_{u,v} \ln(\mathcal{P}[u, v] + \varepsilon)\right)}{\frac{1}{HW} \sum_{u,v} \mathcal{P}[u, v]}, \quad (16)$$

where $\varepsilon > 0$ prevents $\ln(0)$. The numerator is the geometric mean and the denominator is the arithmetic mean; by the AM–GM inequality, $\gamma \in (0, 1]$, with $\gamma = 1$ for spectrally flat (white noise) and $\gamma \rightarrow 0$ for a single pure tone.

Proposition 4.1. AI-generated images exhibit γ closer to unity than natural images of equivalent content, because the generative synthesis process distributes energy more uniformly across spatial frequencies.

4.3 High-to-Low-Frequency Energy Ratio

Partition the frequency plane at threshold radius $\rho_0 = 0.3 R_{\max}$:

$$E_{\text{LF}} = \frac{1}{|\Omega_{\text{in}}|} \sum_{(m,n): \rho[m,n] < \rho_0} \mathcal{P}[m, n], \quad (17)$$

$$E_{\text{HF}} = \frac{1}{|\Omega_{\text{out}}|} \sum_{(m,n): \rho[m,n] \geq \rho_0} \mathcal{P}[m, n], \quad (18)$$

$$\lambda = \frac{E_{\text{HF}}}{E_{\text{LF}}}. \quad (19)$$

For natural images $\lambda \ll 1$ (due to the $1/f^2$ law); elevated λ indicates anomalous high-frequency energy injection consistent with generative up-sampling.

4.4 Complete Feature Vector

The final feature representation is the concatenation:

$$\mathbf{x} = (\phi_{00}, \dots, \phi_{(N_r-1)(N_s-1)}, \eta_{\text{DC}}, \gamma, \lambda) \in \mathbb{R}^{N_r N_s + 3} = \mathbb{R}^D. \quad (20)$$

Prior to classification, each component is z-score standardised: $\tilde{x}_k = (x_k - \mu_k)/\sigma_k$ over the training corpus.

4.5 Grid Artefact Score

To directly quantify the presence of periodic spectral peaks (Section 3.5), we define a residual spectrum by suppressing the DC lobe at radius ρ_{dc} :

$$\mathcal{L}_{\text{res}}[m, n] = \begin{cases} 0 & \rho[m, n] \leq \rho_{\text{dc}}, \\ \mathcal{L}[m, n] & \text{otherwise,} \end{cases} \quad (21)$$

and counting high-energy residual pixels above the τ -th quantile:

$$\mathcal{K} = |\{(m, n) \mid \mathcal{L}_{\text{res}}[m, n] > Q_\tau(\mathcal{L}_{\text{res}})\}|. \quad (22)$$

We use $\tau = 0.80$. Empirically, $\mathcal{K} > 0.01 HW$ is a reliable heuristic for AI-generated content.

5. Detection Pipeline

Algorithm 1 FFT-Based AI Image Detection

Require: Image $\mathbf{I}_{\text{color}} \in \mathbb{R}^{H \times W \times 3}$

Ensure: Classification $\hat{y} \in \{0, 1\}$, confidence p

- 1: **Preprocess:** $\mathbf{I} \leftarrow \text{GRAYSCALE}(\mathbf{I}_{\text{color}})$; re-size to 128×128
 - 2: **FFT:** $F \leftarrow \mathcal{F}_{2D}(\mathbf{I})$
 - 3: **Shift:** $\hat{F}[u, v] \leftarrow F[u - \frac{H}{2}, v - \frac{W}{2}]$
 - 4: **Log magnitude:** $\mathcal{L}[u, v] \leftarrow 20 \log(1 + |\hat{F}[u, v]|)$
 - 5: **Extract features:**
 - 6: $\{\phi_{ij}\} \leftarrow$ polar bin averages via (15)
 - 7: $\eta_{\text{DC}} \leftarrow$ DC fraction via (11)
 - 8: $\gamma \leftarrow$ spectral flatness via (16)
 - 9: $\lambda \leftarrow$ HF/LF ratio via (19)
 - 10: **Assemble:** $\mathbf{x} \leftarrow (\phi_{ij}, \eta_{\text{DC}}, \gamma, \lambda) \in \mathbb{R}^{515}$
 - 11: **Standardise:** $\tilde{\mathbf{x}} \leftarrow (\mathbf{x} - \boldsymbol{\mu}) / \boldsymbol{\sigma}$
 - 12: **Classify:** $(\hat{y}, p) \leftarrow \mathcal{C}(\tilde{\mathbf{x}}) \quad \triangleright$ Trained RF / LR / GBT
-

5.1 Training Data

We leverage two publicly available corpora:

Natural images. The CIFAR-10 dataset [12] provides 60,000 real-world photographs across 10 object categories. Despite the modest native resolution (32 px), upscaling to 128×128 preserves the characteristic $1/f^2$ spectral structure of photographic imagery.

AI-generated images. DiffusionDB [13] contains over 14 million images generated by Stable Diffusion [1] from diverse text prompts. We use the `2m_first_1k` subset, which requires no authentication and streams directly via the HuggingFace `datasets` API.

5.2 Classifiers

Three classifiers are evaluated on the standardised feature vectors:

Logistic Regression (LR)

$P(\hat{y} = 1 \mid \tilde{\mathbf{x}}) = \sigma(\mathbf{w}^\top \tilde{\mathbf{x}} + b)$, trained by minimising L_2 -regularised cross-entropy loss.

Random Forest (RF)

An ensemble of $T = 200$ trees with bootstrap aggregation. Split quality is assessed via the Gini impurity criterion:

$$\text{Gini}(\mathcal{S}) = 1 - \sum_{c \in \{0,1\}} p_c^2. \quad (23)$$

RF additionally provides per-feature importance scores (mean decrease in Gini impurity), enabling post-hoc interpretability of which spectral regions are most discriminative.

Gradient Boosting (GBT)

An additive model $F_T = \sum_{t=1}^T \nu h_t$ where each shallow tree h_t fits the negative gradient (pseudo-residuals) of the log-loss:

$$r_i^{(t)} = y^{(i)} - \sigma(F_{t-1}(\tilde{\mathbf{x}}^{(i)})). \quad (24)$$

6. Evaluation Methodology

6.1 Metrics

Performance is evaluated via the *Area Under the ROC Curve* (AUC):

$$\text{AUC} = P(s(\mathbf{x}^+) > s(\mathbf{x}^-)), \quad (25)$$

where $\mathbf{x}^+$ is a randomly drawn AI image and $\mathbf{x}^-$ a natural image. This metric is threshold-independent and handles class imbalance without modification. We also report precision, recall, and F_1 score at the optimal operating point $\delta^* = \arg \min_{\delta} |P(\delta) - R(\delta)|$.

6.2 Cross-Validation

Generalisation is estimated via 5-fold stratified cross-validation on the training split. The mean and standard deviation of per-fold AUC provide uncertainty quantification:

$$\widehat{\text{AUC}}_{\text{CV}} \pm \widehat{\sigma}_{\text{CV}} = \frac{1}{5} \sum_{k=1}^5 \text{AUC}_k \pm \sqrt{\frac{1}{4} \sum_{k=1}^5 (\text{AUC}_k - \bar{\text{AUC}})^2} \quad (26)$$

7. Theoretical Analysis

7.1 Why the FFT Reveals Generator Artefacts

The core insight is that any convolutional operation with spatial periodicity leaves a *multiplicative* signature in the Fourier domain. Formally, convolution in space corresponds to pointwise multiplication in frequency:

$$(I * k)[m, n] \xleftrightarrow{\mathcal{F}} F_I[u, v] \cdot K[u, v], \quad (27)$$

where k is a convolutional kernel and $K = \mathcal{F}(k)$. If k has periodic structure (as occurs in strided convolution or learned upsampling filters), K will contain large-magnitude entries at periodic frequency locations. These entries amplify the corresponding components of F_I , producing the characteristic grid of spectral peaks.

7.2 Robustness to Post-Processing

Mild JPEG compression retains spectral peaks because JPEG operates on 8×8 DCT blocks — it smooths high-frequency components within blocks but does not destroy the inter-block periodic structure introduced by generator strides. Resizing to 128×128 similarly preserves the relative spectral shape. The principal threat to robustness is adversarial post-processing (e.g., targeted spectral smoothing), which we consider a direction for future work.

7.3 Connection to the $1/f$ Law

The $1/f^2$ power spectrum of natural images arises from the *scale-invariance* of natural scenes [11]: objects at different scales appear statistically similar, leading to self-similar statistics in which power scales as the inverse square of frequency. Generative models break this invariance because they synthesise images at a fixed resolution from a latent space of finite dimensionality, without an explicit mechanism to enforce scale-invariant statistics across all frequencies simultaneously.

8. Limitations and Future Directions

Several important limitations should be acknowledged. First, the feature vector is computed from a single grayscale channel; extending to per-channel or inter-channel cross-spectral analysis may improve sensitivity to chromatic artefacts introduced by colour-space operations in VAE decoders. Second, adversarial attackers aware of the spectral detection strategy could apply targeted frequency-domain smoothing, selectively suppressing the diagnostic peaks while preserving perceptual quality. Third, the $1/f^2$ baseline is an ensemble property of natural image statistics; individual photographs (e.g., high-contrast graphic art, medical imaging) may deviate from this law, potentially increasing false-positive rates.

Promising future directions include: (i) adversarial training of the classifier against spectrally-aware image perturbations; (ii) extension to video sequences, where temporal frequency analysis adds an additional discriminative dimension; (iii) cross-architecture generalisation studies using held-out generators not seen during training.

9. Conclusion

We have presented a frequency-domain framework for detecting AI-generated images grounded in the statistical physics of natural image spectra. The central thesis is that generative architectures — regardless of their specific design — must perform spatial-frequency operations (upsampling, strided convolution, attention tiling) that leave measurable, persistent signatures in the 2-D Fourier spectrum. These signatures manifest as three characteristic deviations from natural image statistics: suppressed DC bias ($\eta_{\text{DC}} \downarrow$), elevated spectral flatness ($\gamma \uparrow$), and periodic grid-like magnitude peaks.

A 515-dimensional polar feature vector extracted from the log-magnitude spectrum captures these deviations compactly and feeds a family of interpretable classifiers. The pipeline

is lightweight, requires no GPU inference, and connects seamlessly to open image libraries for reproducible training and evaluation.

As generative models continue to advance, frequency-domain forensics offers a stable, principled foundation: the laws of spectral statistics are architectural constraints, not learned behaviours, and they cannot be circumvented without fundamentally altering the synthesis process itself.

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